CS24351 — ELECTIVE (3.0 CR) | MODULE II NOTES | BIT MESRA | NEP SCHEME

Module 2: Language Models, Smoothing & Parsing

N-Grams, Perplexity, Laplace/Kneser-Ney, HMM Viterbi, PCFG

Module II: Language Models, Smoothing & Parsing — Topics Covered

1. N-gram Language Models & Markov Assumption
2. Perplexity Evaluation Metric
3. Smoothing: Laplace, Good-Turing, Kneser-Ney
4. HMM POS Tagging & The Viterbi Algorithm
5. Probabilistic Context-Free Grammars (PCFGs)

1. N-gram Language Models & Perplexity

$$P(w_1, w_2, \ldots, w_N) = \prod_{k=1}^{N} P(w_k \mid w_{k-n+1}^{k-1})$$

$$\text{Perplexity}(W) = P(w_1, \ldots, w_N)^{-1/N} = 2^{-\frac{1}{N}\sum \log_2 P(w_i|w_{i-n+1}^{i-1})}$$

2. Smoothing Techniques Comparison

- **Laplace (Add-1) Smoothing:** $P(w_n \mid w_{n-1}) = \frac{C(w_{n-1}, w_n)+1}{C(w_{n-1})+V}$. (Tends to allocate too much probability mass to unseen items).
- **Kneser-Ney Smoothing:** State-of-the-art N-gram smoothing combining absolute discounting with continuation probabilities based on how versatile a word is as a novel continuation.